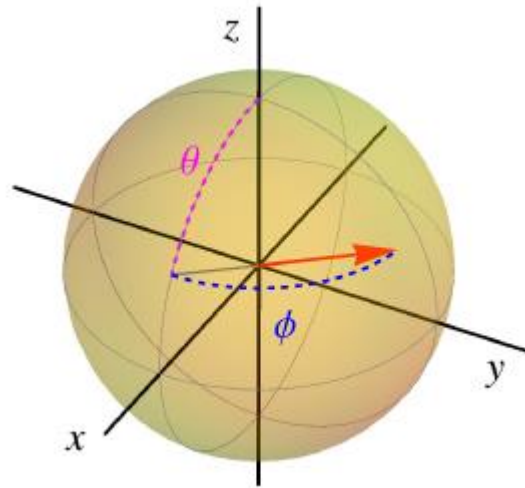


Quantum Computation

ELK336E

Week #8



PHASE ESTIMATION

The eigenvalues of unitary operators are phases.

$$U|\psi_\lambda\rangle = e^{+i\lambda}|\psi_\lambda\rangle$$

$$\lambda = 2\pi\varphi$$

$$0 \leq \varphi \leq 1 \quad \varphi = 0.x_1x_2x_3 \dots$$

$$U|\phi\rangle = e^{2\pi i\varphi}|\phi\rangle$$

$$e^{2\pi i \varphi} = e^{2\pi i 0.x_1 x_2 x_3}$$

$$e^{2\pi i 2^1 \varphi} = e^{2\pi i x_1.x_2 x_3} = \underbrace{e^{2\pi i x_1}}_1 e^{2\pi i 0.x_2 x_3}$$

$$e^{2\pi i 2^2 \varphi} = e^{2\pi i x_1 x_2 x_3} = \underbrace{e^{2\pi i x_1 x_2}}_1 e^{2\pi i 0.x_3}$$

$$\vdots$$

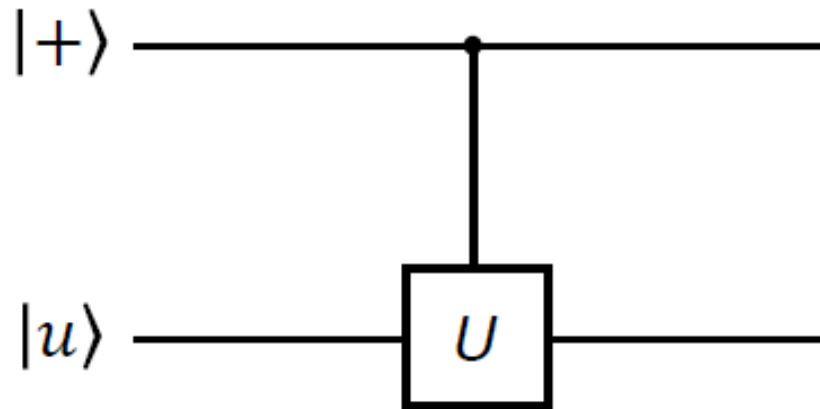
$$e^{2\pi i 2^{(j-1)} \varphi} = e^{2\pi i 0.x_j x_{j+1}}$$

Each power of the operator extracts one more digit.

$$e^{2\pi i \phi 2^k} |\phi\rangle = (e^{2\pi i \phi})^{2^k} |\phi\rangle \\ = U^{2^k} |\phi\rangle$$

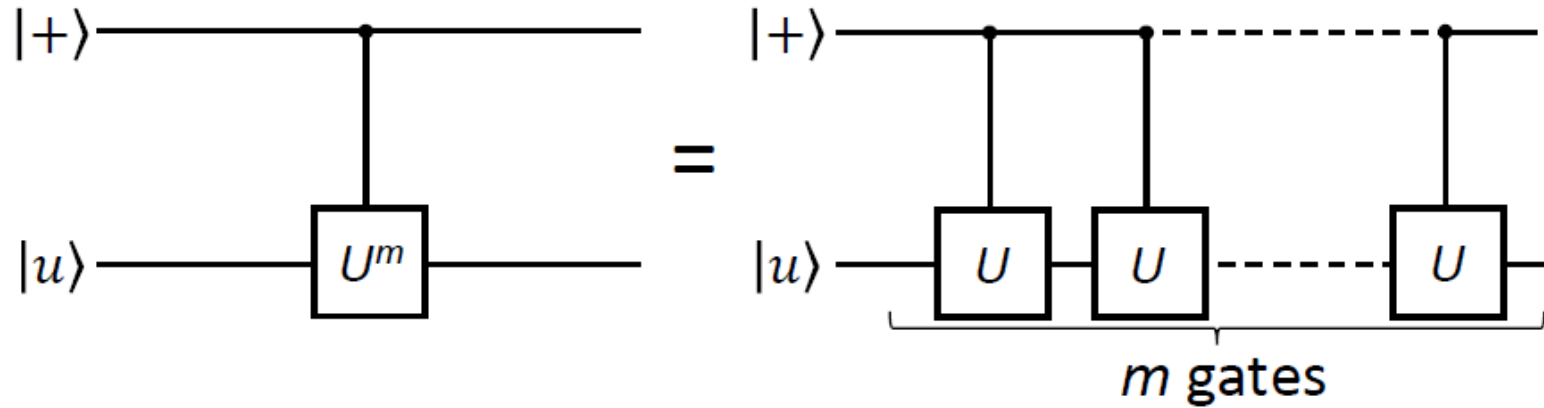
We should apply each step of the algorithm in a controlled fashion.

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) |u\rangle \rightarrow \frac{1}{\sqrt{2}} (|0\rangle |u\rangle + |1\rangle e^{2\pi i \phi} |u\rangle)$$

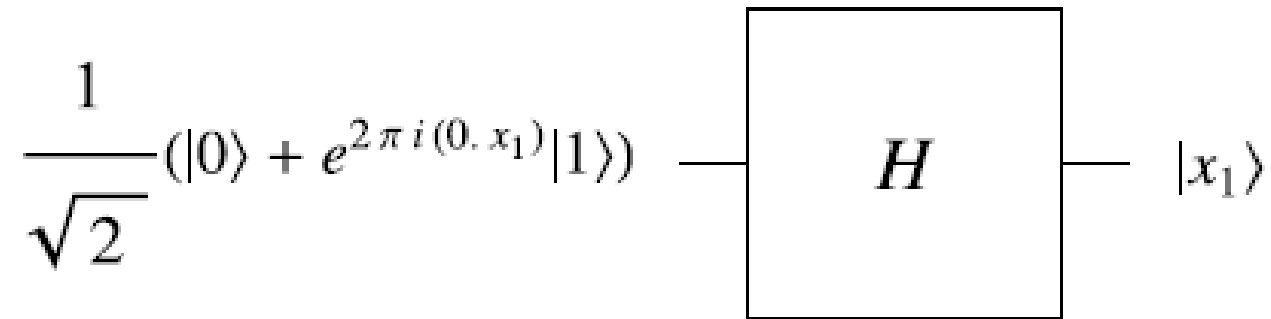
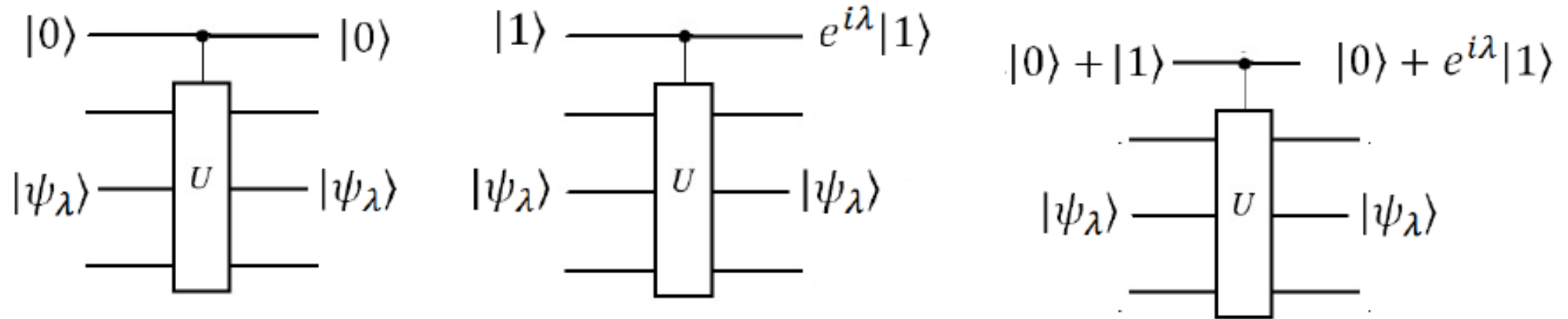


$$= \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i \phi} |1\rangle) |u\rangle$$

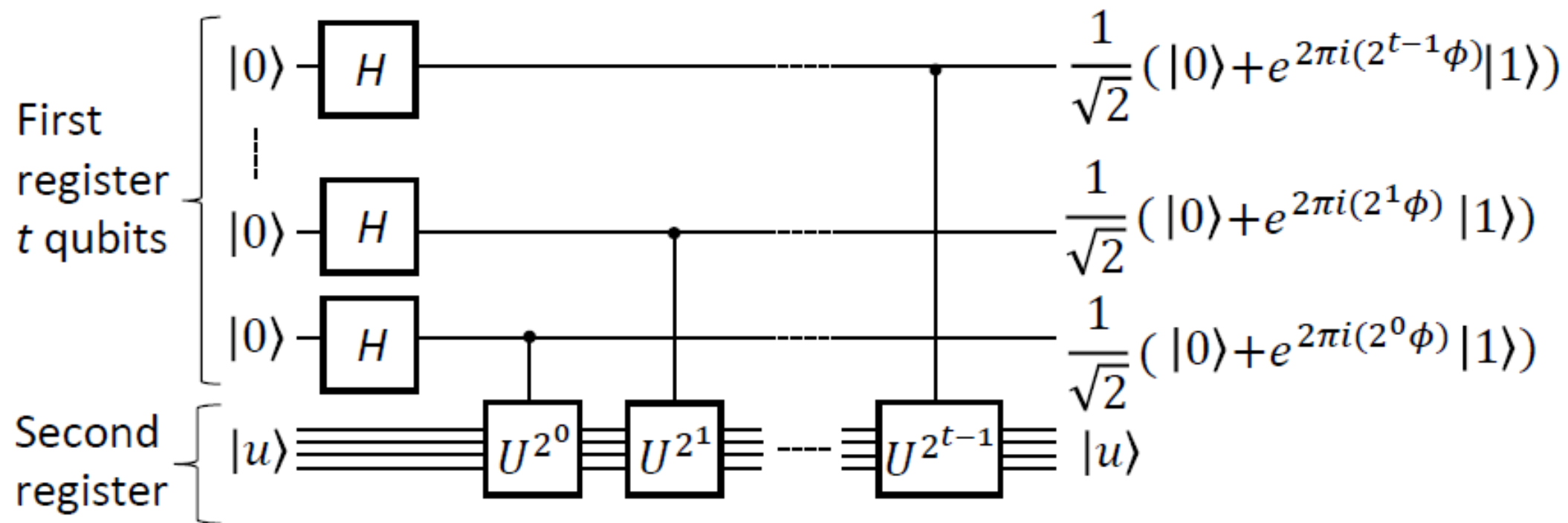
In QPE we have to apply U repeatedly:

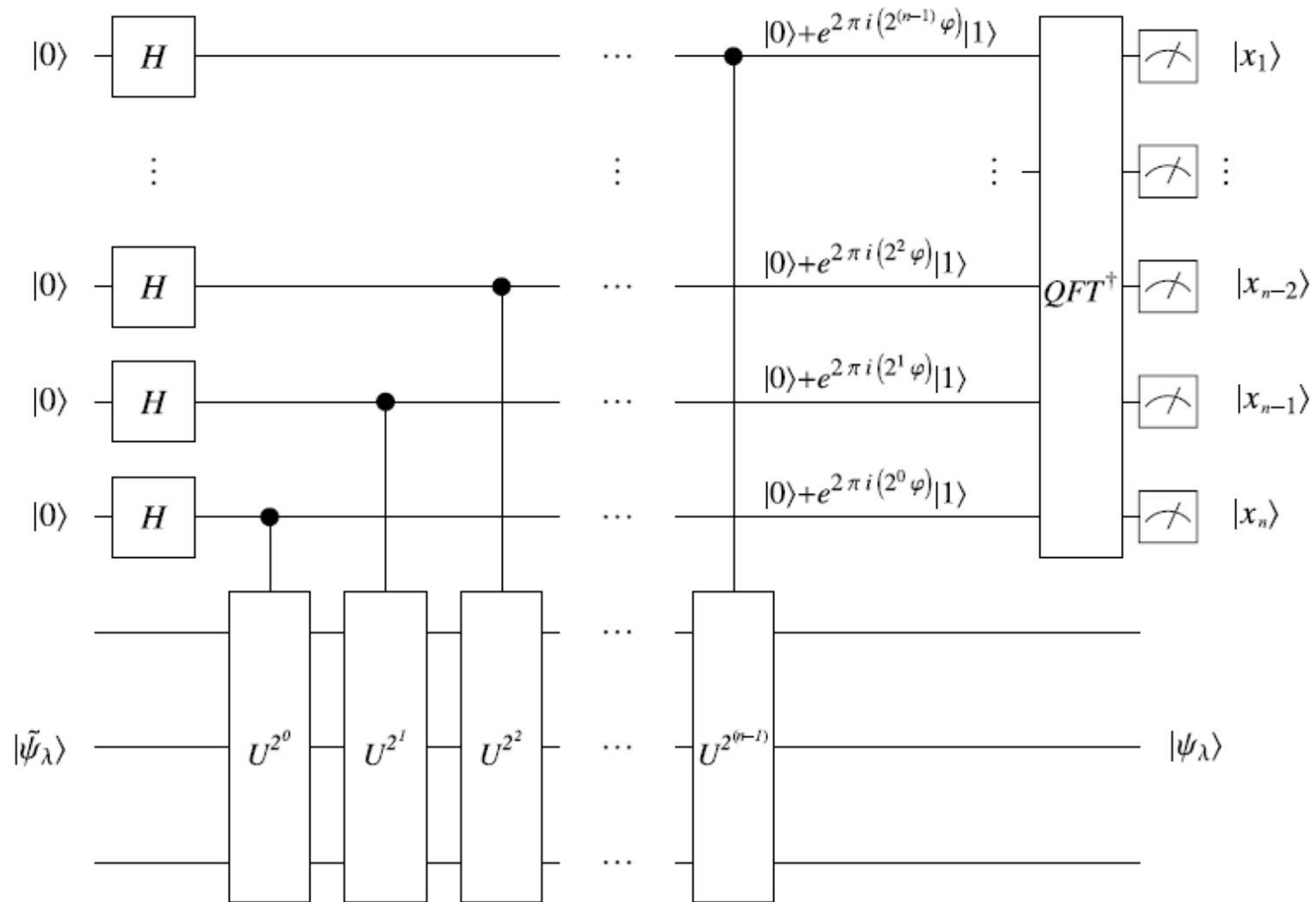


$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) |u\rangle \rightarrow \frac{1}{\sqrt{2}} (|0\rangle + e^{2m\pi i\phi} |1\rangle) |u\rangle$$



One can extract the phase information by applying an inverse H gate to the first register.





Abrams-Lloyd Eigenvalue Estimation Algorithm.

QFT, Reminder and a Two-Qubit example

$$J_1 J_2 \dots J_n = 2^n (0.J_1 J_2 \dots J_n)$$

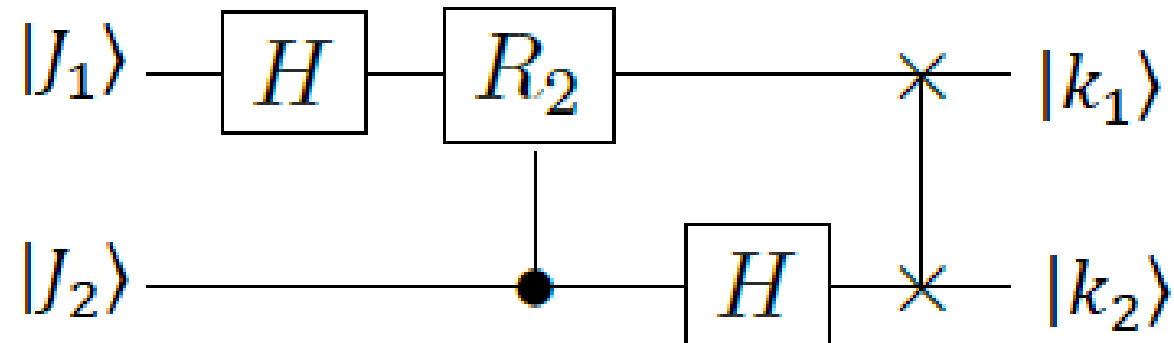
$$0.J_1 J_2 \dots J_n = 0.J_1 + 0.0J_2 + 0.00J_3 + \dots$$

$$= J_1 2^{-1} + J_2 2^{-2} + J_3 2^{-3} + \dots$$

$$0.J_1 J_2 \dots J_n = \sum_k J_k 2^{-k}$$

$$QFT|J_1J_2\cdots J_n\rangle = \left(\frac{|0\rangle + e^{2\pi i 0.J_n}|1\rangle}{\sqrt{2}}\right) \otimes \cdots \otimes \left(\frac{|0\rangle + e^{2\pi i 0.J_1J_2\cdots J_n}|1\rangle}{\sqrt{2}}\right)$$

EXAMPLE:



In this two-qubit QFT example ;

(a) Find the $|k_1k_2\rangle$

(b) Obtain the matrix representation of the circuit.

Solution :

Remind that $H|j\rangle = \frac{1}{\sqrt{2}} \sum_{j \in \{0,1\}} (-1)^j |j\rangle$

This leads to $H|0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$ and $H|1\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$

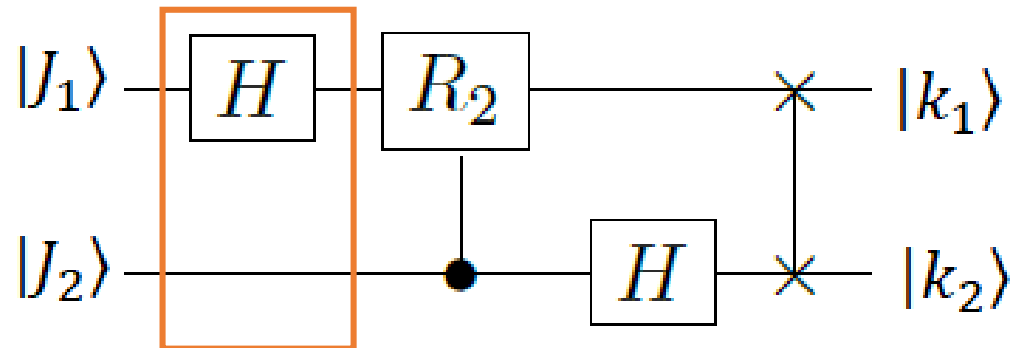
Hadamard can also be expressed as $H|j\rangle = \frac{1}{\sqrt{2}} \sum_{j \in \{0,1\}} e^{2\pi i 0 \cdot j} |j\rangle$

$$|j\rangle \xrightarrow{H} \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i 0 \cdot j} |1\rangle)$$

Also remember that $R_k = \begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{bmatrix}$

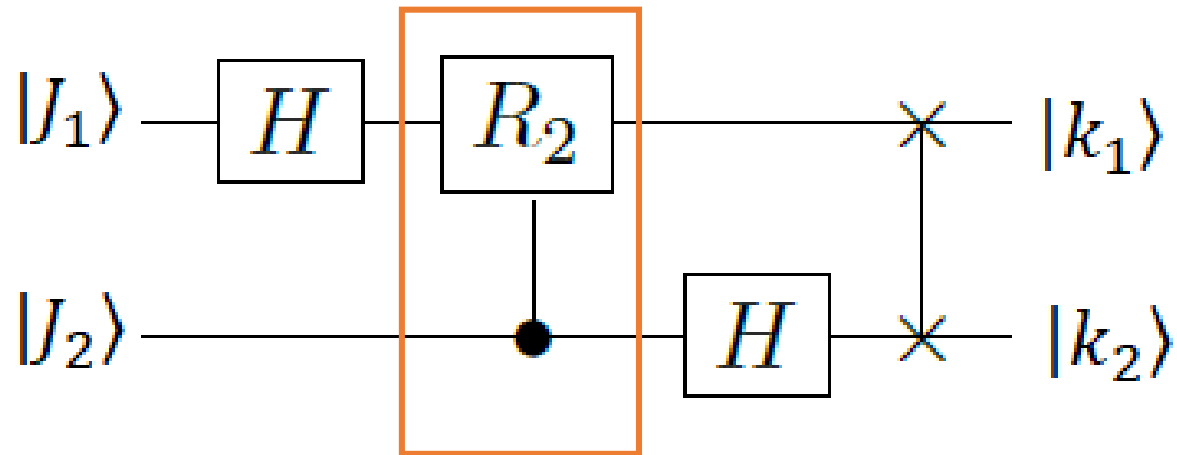
$$R_k |0\rangle = |0\rangle, \quad R_k |1\rangle = e^{2\pi i/2^k} |1\rangle$$

$$|J_1 J_2\rangle \xrightarrow{H \otimes \mathbb{1}} \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i 0 \cdot J_1} |1\rangle) |J_2\rangle = \frac{1}{\sqrt{2}} (|0\rangle |J_2\rangle + e^{2\pi i 0 \cdot J_1} |1\rangle |J_2\rangle)$$

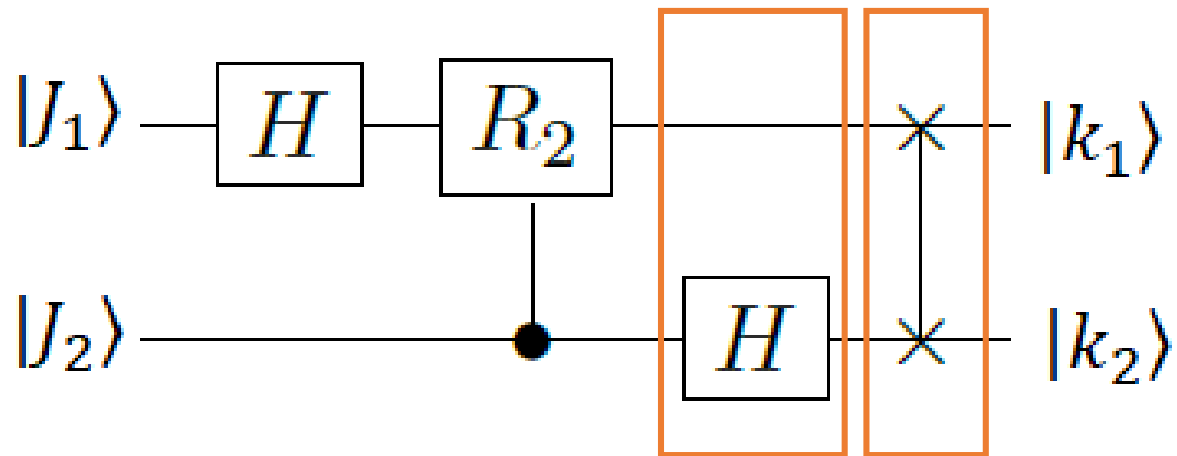


$$\xrightarrow{[C-R_2]^{21}} \frac{1}{\sqrt{2}} (|0\rangle|J_2\rangle + e^{2\pi i 0.J_1} e^{2\pi i 0.0J_2} |1\rangle|J_2\rangle)$$

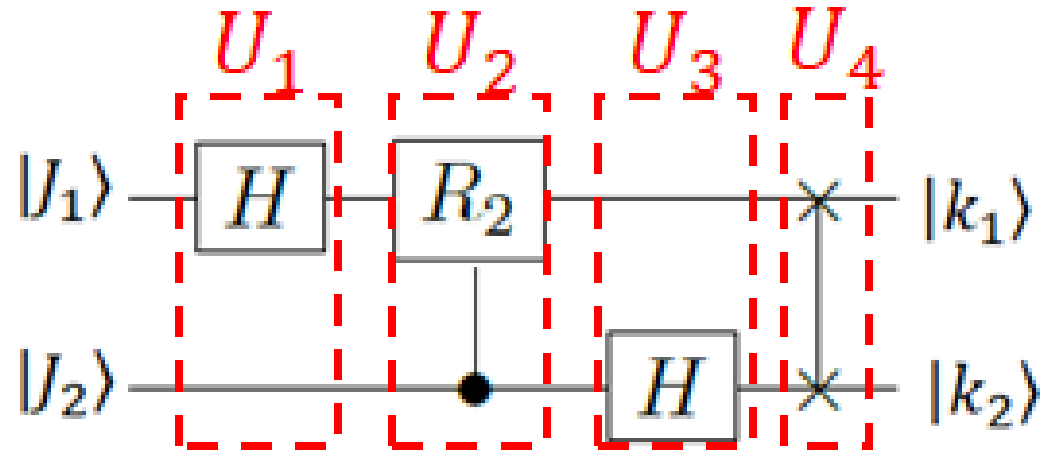
$$= \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i 0.J_1J_2} |1\rangle)|J_2\rangle)$$



$$\xrightarrow{1 \otimes H} \left(\frac{|0\rangle + e^{2\pi i 0 \cdot J_1 J_2} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + e^{2\pi i 0 \cdot J_2} |1\rangle}{\sqrt{2}} \right)$$



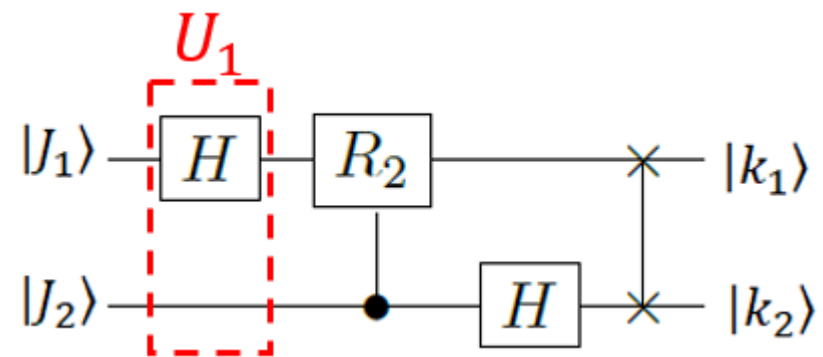
$$\xrightarrow{SWAP} \left(\frac{|0\rangle + e^{2\pi i 0 \cdot J_2} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + e^{2\pi i 0 \cdot J_1 J_2} |1\rangle}{\sqrt{2}} \right) = |k_1 k_2\rangle$$



Matrix representation:

$$|k_1 k_2\rangle = U_4 U_3 U_2 U_1 |J_1 J_2\rangle$$

$$QFT_2 = U_4 U_3 U_2 U_1$$



$$U_1 = H \otimes \mathbb{1} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$U_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$

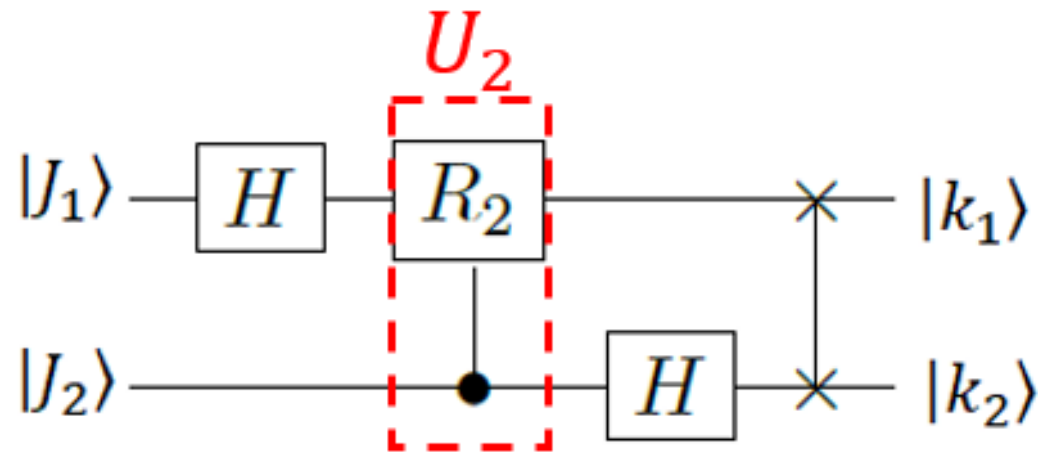
$$|0\rangle|0\rangle \rightarrow |0\rangle|0\rangle$$

$$|0\rangle|1\rangle \rightarrow (R_2|0\rangle)|1\rangle = |0\rangle|1\rangle$$

$$|1\rangle|0\rangle \rightarrow |1\rangle|0\rangle$$

$$|1\rangle|1\rangle \rightarrow (R_2|1\rangle)|1\rangle = e^{i\frac{\pi}{2}}|1\rangle|1\rangle$$

$$U_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\frac{\pi}{2}} \end{bmatrix}$$

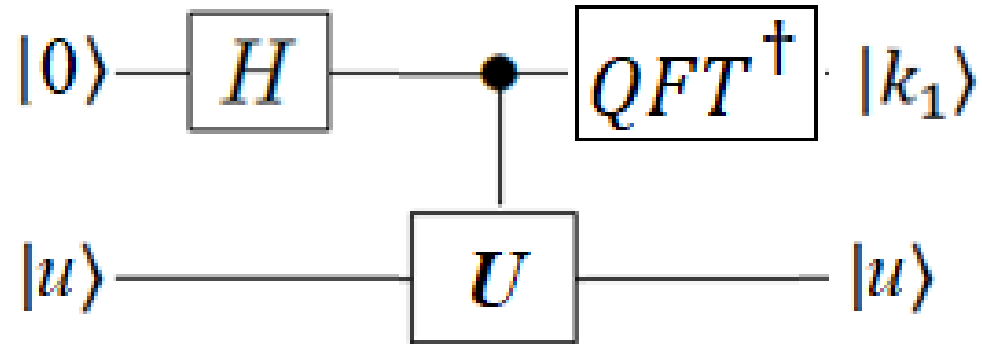


$$U_3 = \mathbb{1} \otimes H = \begin{bmatrix} H & 0 \\ 0 & H \end{bmatrix} \quad U_3 = \mathbb{1} \otimes H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$QFT_2 = U_4 U_3 U_2 U_1$$

$$QFT_2 = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & -i \end{bmatrix}$$

EXAMPLE:



Let, $U = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ and $|u\rangle = \begin{bmatrix} 1 \\ -i \end{bmatrix}$,
show that $|k_1\rangle = |1\rangle$.

Remember that the condition is ;

$$U|u\rangle = e^{2\pi i\varphi}|u\rangle$$

$$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} -1 \\ i \end{bmatrix}$$

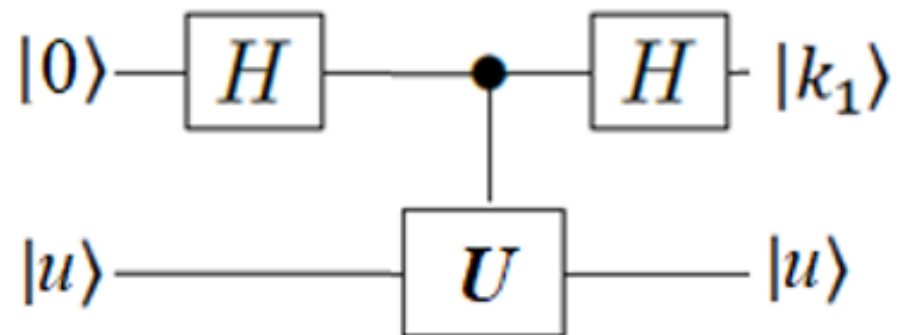
$$= - \begin{bmatrix} 1 \\ -i \end{bmatrix} = -|u\rangle$$

$$e^{2\pi i\varphi} = -1$$

$$\varphi = \frac{1}{2} = 2^{-1} = (0.1)_2 = 0.k_1$$

One qubit quantum Fourier transform is the Walsh Hadamard transform; therefore;

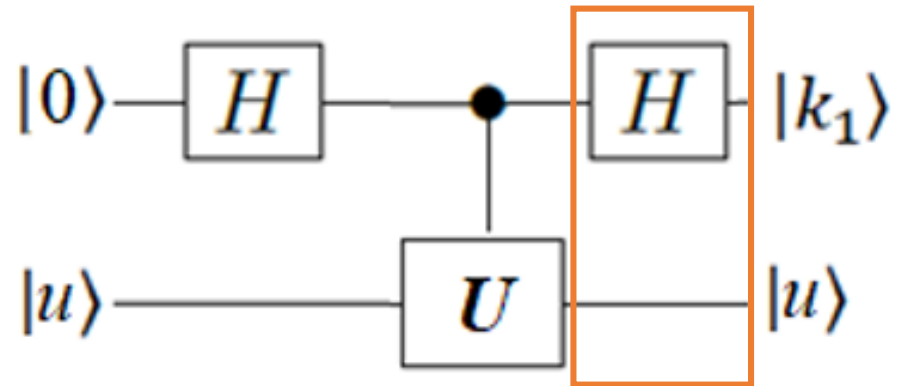
$$QFT_1 = QFT_1^\dagger = H$$



$$|0\rangle|u\rangle \xrightarrow{H \otimes 1} \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) |u\rangle$$

$$\xrightarrow{C-U} \frac{1}{\sqrt{2}} (|0\rangle|u\rangle + |1\rangle U|u\rangle)$$

$$= \frac{1}{\sqrt{2}} (|0\rangle|u\rangle - |1\rangle|u\rangle) = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) |u\rangle$$



$$\xrightarrow{H \otimes \mathbb{1}} \frac{1}{2} \left((|0\rangle + |1\rangle) - (|0\rangle - |1\rangle) \right) |u\rangle$$

$$= |1\rangle |u\rangle \quad k_1 = 1$$